Ali Abdari

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PhD (2025) in Artificial Intelligence with 6+ years of experience developing and deploying deep learning systems for image and video understanding. Expertise in real-time video segmentation, multi-object tracking, action recognition in raw and compressed video, and multimodal vision-language models. Experienced in adapting, fine-tuning, and profiling large foundation models, including ViT-L-based segmentation models for latency-constrained inference in PyTorch, ONNX, and TensorRT. Proven track record of translating research into production systems in healthcare and automotive, and publishing at ICMR, MMM, IEEE TMM, and ACM TOMM.

Professional Experience

Computer Vision and AI Researcher — SMC | a DGS Company 2026 – Present

- **Real-Time Medical Image/Video Segmentation and Tracking (Hospital of Padua):** Fine-tuned SAM 3 (ViT-L) with LoRA and SegFormer for surgical and clinical footage; profiled the inference pipeline stage by stage and accelerated it 4× with ONNX and TensorRT; integrated the models into CVAT to semi-automate annotation.
- **3D Deep Learning for Automotive (Stellantis):** Evaluating B-rep, mesh, and point-cloud deep learning models for automotive geometry tasks; benchmarking representation trade-offs for accuracy and inference cost.
- **LLM-Based Report Generation:** Built an agentic system that answers natural-language questions over company databases and generates reports; deployed to multiple clients.
- **Warehouse Layout Optimization:** Applying simulated annealing and genetic algorithms to minimize item retrieval cost.

Computer Vision Engineer — Caica 2022 – 2023

- **Primate Behavior Analysis via Video** (German Primate Center, DPZ)**:** Developed a system for object detection, multi-object tracking, keypoint detection, and behavioral analysis; fine-tuned YOLO models and used DeepSORT for tracking.
- **Fatigue Detection System Using Facial Keypoints:** Developed a fatigue detection system using MediaPipe to detect facial keypoints and measure eye blinks; deployed for factory workers to improve workplace safety.
- **AI-Based Assistive System for Patients Using Eye Tracking:** Developed an assistive system using MediaPipe to track eye direction and detect eye blinks, enabling patients to type on a virtual keyboard using eye movements.

Computer Vision Engineer — Iran's National Elites Foundation 2019 – 2021

- **Traffic and Pedestrian Video Synopsis:** Analyzed movements in urban and non-urban areas using DeepSORT and FairMOT; detected abnormal traffic patterns; generated concise summaries of long videos by temporally rearranging moving-object trajectories, reducing video length to about 1/5 of the original for efficient review.

Android and Python Developer — Mosbat Company 2018 – 2020

Python Developer — Vestaak Company 2017 – 2018

Education

Ph.D. in Artificial Intelligence – University of Naples Federico II, University of Udine, Italy 2023 – 2025
Thesis area: Complex 3D Scene and Metaverse Retrieval Using Natural Language

M.S. in Computer Science – Kharazmi University, Iran 2016 – 2019
Thesis area: Action Recognition in Compressed Video

B.S. in Computer Engineering – Kharazmi University, Iran 2012 – 2016

Skills

Architectures — Vision Transformers (ViT, ViTDet), SAM 3 / SAM family, DETR-style detectors, SegFormer, CLIP-style contrastive vision-language models, YOLO, LoRA fine-tuning, 3D point-cloud encoders

Optimization & Deployment — ONNX, TensorRT, inference profiling, FlashAttention, torch.compile, mixed precision, CVAT, Docker, Kubernetes, AWS, Azure ML

ML/DL Frameworks — PyTorch, TensorFlow, Keras, Hugging Face, OpenCV, Dlib, MediaPipe, NumPy, Pandas, Matplotlib, scikit-learn

Programming Languages — Python, C++, C, MATLAB, SQL

Languages — English (C1, IELTS 7.5), Italian (C1), Persian (Native)

Academic Activities

Organization — CV4Metaverse Workshop at ECCV 2024, Italy and CVPR 2025, USA 2024, 2025

Supervision — a Ph.D. student and a research fellow on ML for agricultural and genomic data 2024 – 2025

Organization — Metaverse Apartment Retrieval Challenge at CV4Metaverse, ECCV 2024, Italy 2024

Teaching Assistant — Deep Learning, University of Udine 2023 – 2024

Computer Vision and Machine Learning Researcher — Kharazmi University 2021 – 2023

Teaching Assistant — Seven computer science courses (B.S. and M.S.), Kharazmi University 2013 – 2019

Selected Publications (Google Scholar)

1. **Abdari, A.**, Falcon, A., & Serra, G. (2026). Retrieving Relevant Metaverses Using Hierarchical Features. ACM Transactions on Multimedia Computing, Communications and Applications.
2. Falcon, A., **Abdari, A.**, & Serra, G. (2025). ALCER3D: Adaptive Learning Constraints for Enhanced Retrieval of Complex Indoor 3D Scenarios. IEEE Transactions on Multimedia.
3. **Abdari, A.**, Falcon, A., & Serra, G. (2024). A Language-Based Solution to Enable Metaverse Retrieval. In International Conference on Multimedia Modeling (MMM). Springer Nature Switzerland.
4. **Abdari, A.**, Falcon, A., & Serra, G. (2024). AdOCTeRA: Adaptive Optimization Constraints for Improved Text-Guided Retrieval of Apartments. In Proceedings of the 2024 International Conference on Multimedia Retrieval (ICMR).
5. Honarjoo, N., **Abdari, A.**, & Mansouri, A. (2024). Violence Detection in Compressed Video. Multimedia Tools and Applications.